

LN8810

Version: 1.03

Date:2019-03-06

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Revision History

Version	Date	Author	Summary
V1.01	2018-12-26		Draft
V1.02	2019-01-30	Jack	Change the boot configure option
V1.03	2019-03-06	Jack	Add RF performance

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1 Overview

1.1 Introduction

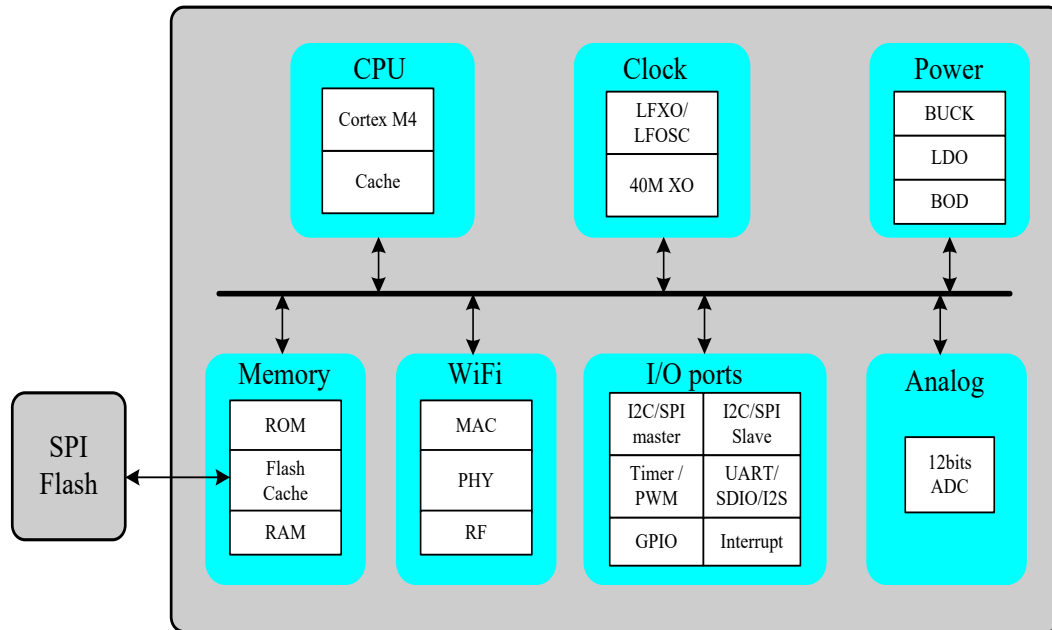
The chip integrates WiFi RF transceiver, baseband and MAC. An ARM® Cortex M4F embedded 32-bits MCU is integrated for IP stack and applications.

LN8810 is a highly integrated Wi-Fi® Chip targeted for various IOT applications that can benefit from its good RF performance, low-power consumption and easy application development, which means fast time to market for IOT products. LN8810 integrates all the necessary elements required for IOT Wi-Fi application, including an 802.11b/g/n radio, a micro-controller, Wi-Fi and IP stacks, an HTTP server, and multiple protocols, such as TCP/IP and UDP. LN8810 works as a Wi-Fi client or access point, making the device connected online very easily, and applications can be designed without an external MCU. Alternatively, it also has highly flexible hardware interfaces which allows connection to different peripherals and sensors.

1.2 Features

- ✧ Main Features
 - 2.4GHz 802.11 b/g/n
 - Support STA/AP/STA+AP operation modes
 - Support WPA/WPA2
 - 18dBm output power in 11b mode
 - -94dBm sensitivity
- ✧ High Integration
 - Integrated ARM® Cortex M4F
 - Integrated TCP/IP protocol stack
 - Integrated switch, balun, LNA, power amplifier
 - High precision oscillator
 - Integrated regulators to support single supply voltage
- ✧ Interface
 - Single supply voltage with wide range supported (2.5v~3.6V)
 - SPI/SDIO/UART host interface
 - SPI/I2C master for sensor
 - I2S, PWM and GPIO
 - Multi-channel ADC input
- ✧ Low Power
 - Active mode :60mA in RX mode, 220mA in TX mode @3.3V
 - 1.5mA in standby mode (DTIM=1)
 - 250uA in sleep mode (fast wakeup)
 - 50uA in deep sleep mode
- ✧ MCU
 - 80/160MHz clock for M4
 - 256KB on-chip SRAM
 - Up to 2MB Flash for application code
 - Program can run directly from SPI flash
 - SWD debug interface
- ✧ Package
 - QFN 48 6mm*6mm*0.75mm

1.3 Block diagram



1.4 Application

With low power and low cost, LN8810 will speed up things keeping online everywhere with real wireless connection powered by battery. And so, electric tail is not necessary.

- Lower power control, access and data collection: with ultra-low standby power consumption, it can keep the things online anywhere with battery supplied
- Wireless audio: very low receiving power for longer battery life and high throughput for audio quality
- Wireless video: With low power and low cost, those toy drones, cars can also achieve real time video transferring by LN8810

2 Pin Description

QFN 48	Name	Type	Description	Second function	Third function	Fourth function
1	NC	NC	Non-Connection			
2	RFIO	IO	RF IO for antenna			
3	NC	NC	Non-Connection			
4	RESETN	DI	Chip reset input, low active with internal pull-up resistor			
5	GPIO20	DIO	Digital general purpose In and Output with interrupt function	spim_miso	fullmux[12]	
6	GPIO19	DIO	Digital general purpose In and Output with interrupt function	spim_mosi	fullmux[11]	
7	GPIO29	DIO	Digital general purpose In and Output with interrupt function			
8	GPIO27	DIO	Digital general purpose In and Output with interrupt function			
9	GPIO32	DIO	Digital general purpose In and Output with interrupt function			
10	GPIO18	DIO	Digital general purpose In and Output	spim_csn0		
11	GPIO17	DIO	Digital general purpose In and Output	spim_clk		
12	GPIO33	DIO	Digital general purpose In and Output			
13	VDDIO	P	Power supply for digital IO			
14	GPIO34	DIO	Digital general purpose In and Output with interrupt function			
15	GPIO9	DIO	Digital general purpose In and Output with interrupt function	sdio_clk	spis_clk	i2s_bclk
16	GPIO10	DIO	Digital general purpose In and Output with interrupt function	sdio_cmd	spis_csn	i2s_lrcclk
17	GPIO11	DIO	Digital general purpose In and Output with interrupt function	sdio_io0		i2s_sdo0
18	GPIO12	DIO	Digital general purpose In and Output with interrupt function	sdio_io1	spis_mosi	i2s_sdo1
19	GPIO13	DIO	Digital general purpose In and Output with interrupt function	sdio_io2		i2s_sdi0
20	GPIO14	DIO	Digital general purpose In and Output with interrupt function	sdio_io3	spis_miso	i2s_sdi1
21	GPIO3	DIO	Digital general purpose In and Output with interrupt function	sw_clk		
22	GPIO31	DIO	Digital general purpose In and Output with interrupt function			
23	GPIO30	DIO	Digital general purpose In and Output with interrupt function			
24	GPIO28	DIO	Digital general purpose In and Output with interrupt function			

QFN 48	Name	Type	Description	Second function	Third function	Fourth function
25	VSS_PMU	G	Ground for PMU			
26	BUCK_SW	PIO	Switch output of BUCK			
27	VDD_PMU	P	Power supply for Chip			
28	EN_CHIP	DI	Chip enable signal, high active 3.3V			
29	GPIO16	DIO	Digital general purpose In and Output with interrupt function			
30	GPIO15	DIO	Digital general purpose In and Output with interrupt function			
31	GPIO22	DIO	Digital general purpose In and Output with interrupt function			
32	GPIO21	DIO	Digital general purpose In and Output with interrupt function			
33	GPIO0	DIO	Digital general purpose In and Output with interrupt function	fullmux[0]		
34	AVDD15	P	Analog Voltage input			
35	GPIO8*	AGPIO	Analog/Digital general purpose In and Out	fullmux[6]	ADC3	
36	GPIO7*	AGPIO	Analog/Digital general purpose In and Out	fullmux[5]	ADC4	
37	DVDD	P	Power supply for core	spim_csn2	fullmux[4]	
38	GPIO6*	AGPIO	Analog/Digital general purpose In and Out	spim_csn1	fullmux[3]	ADC0
39	XTALIN	AI	40M XTAL input			
40	XTALOUT	AIO	40M XTAL output			
41	GPIO4	AGPIO	Analog/Digital general purpose In and Out	fullmux[2]	ADC2	
42	GPIO2	DIO	Digital general purpose In and Output with interrupt function	swd		
43	GPIO1	DIO	Digital general purpose In and Output with interrupt function	fullmux[1]		
44	NC					
45	NC					
46	NC					
47	NC					
48	VDDPA	P	Supply for PA			

Notes:

- 1) GPIO8 can't be pull down to ground by resister;
- 2) GPIO6 used as GPADC which is 1.0Vpp max input range;
- 3) GPIO7 & GPIO8 used as GPADC with 3.0V range;
- 4) Full mux function is as the following table;

function name
i2c0_scl
i2c0_sda
i2c1_scl
i2c1_sda
uart0_tx
uart0_rx
uart0_rts
uart0_cts
uart1_tx
uart1_rx
spim2_sck
spim2_ssn
spim2_sdio
pwm0
pwm1
pwm2
pwm3
pwm4
pwm5
pwm6
pwm7

3 Power manage unit

3.1 Power mode

There are 5 power mode to realize lower power consume flexibly:

- **active:** normal mode. SW gate clock of some Peripherals;
- **sleep:** CPU execute WIFI instruction without 'SLEEPDEEP', and gate itself, other module can still be free running, interrupt can wakeup CPU then exit this mode;
- **Deepsleep:** CPU execute WIFI instruction with 'SLEEPDEEP', all digital part are all clock gated except AWO domain, but do not power down; then LDO voltage can be reduced, clock in AWO will be switched to 32KHz; WIC wakeup CPU, then all clock and power recover to normal state;
- **retention:** CPU execute WIFI instruction with 'SLEEPDEEP', all digital part are all power-down except AWO domain and SRAM; registers inside CPU can be remained, clock in AWO will be switched to 32KHz; WIC wakeup CPU, then all clock and power recover to normal state;

Any low power mode can't turn into another low power mode directly, it must switch to active mode first.

All state config such as clock gate, power gate of one domain should configured before executing WIFI.

Power Save Mode Current

Mode	Condition	Min	Typ	Max	Unit
Power Save Mode	DTIM=1*		1.5		mA
	DTIM=10		0.45		mA

Notes:

(*) Test condition: keep connection with AP.

3.2 Power Domain

Power Domain	Description	Min	Typ.	Max	Unit
VDD_PMU	Supply all the chip except for PA		3.3		V
VDD_PA	Supply PA		3.3		V
AVDD1V5	Supply analog & RTC		1.5		V
DVDD	Supply Core		1.1		V
VDDIO	IO Supply voltage	1.8	2.5	3.3	V

4 RF Performance

4.1 Supported Frequencies

It supports 802.11 2.4GHz frequency band

Table 1. Supported Frequencies

Parameter	Condition	Min	Typ	Max	Unit
Receive frequency range 2.4GHz		2412		2484	MHz

4.2 Receiver Characteristics

Table 2. Receive Characteristics

Parameter	Condition	Min.	Typ.	Max.	Unit
Sensitivity					
11b, 1 DSSS	FER<8%, 1024 bytes		-94		dBm
11b, 11 CCK	FER<8%, 1024 bytes		-87		dBm
11g, 6 OFDM	FER<8%, 1024 bytes		-90		dBm
11g, 54 OFDM	FER<8%, 1024 bytes		-74		dBm
11n, MCS0	FER<8%, 1024 bytes		-88		dBm
11n, MCS7	FER<8%, 1024 bytes		-73		dBm
Maximum input level					
11b	FER<8%, 1024 bytes		-10		dBm
11g	FER<8%, 1024 bytes		-10		dBm
11n	FER<8%, 1024 bytes		-10		dBm
Operating power consumption					
11b, 1 DSSS	-90dBm input		63		mA
11b, 11 CCK	-80dBm input		62		mA
11g, 6 OFDM	-80dBm input		66		mA
11g, 54 OFDM	-70dBm input		66		mA
11n, MCS0	-80dBm input		65		mA
11n, MCS7	-70dBm input		66		mA

Table 3. Transmitter Characteristics

Parameter	Condition	Min.	Typ.	Max.	Unit
Output power					
11b, 1 DSSS	Maximum RMS output power measured		18		dBm
11g, 54 OFDM	Maximum RMS output power measured		13		dBm
11n, MCS7	Maximum RMS output power measured		12		dBm
Power consumption					
11b, 11 CCK			260		mA
11g, 54 OFDM			180		mA
11n, MCS7			160		mA

Output EVM					
11b, 1 DSSS			7		%
11g, 54 OFDM			-25		dB
11n, MCS7			-28		dB

5 Memory Mapping

5.1 Overview

LN8810 integrates ROM, RAM and SPI flash to provide applications with variety requirements of memory

5.2 Memory Mapping

Address range	Destination slave	Notes
0x0~0xffff	64KB ROM	
0x1000_0000~0x103f_ffff	2MB Flash Memory	
0x1fff_0000~0x2002_ffff	224KB SRAM	
0x4000_0000~0x4000_3fff	MAC Register	
0x4000_4000~0x4000_7fff	Baseband Register	
0x4000_8000~0x4000_bfff	PMU register	
0x4000_c000~0x4001_ffff	DMA Register	

0x4001_0000~0x4001_3fff	Flash Controller Register	
0x4001_4000~0x4001_7fff	Cache Controller Register	
0x4001_8000~0x4001_bfff	UART	
0x4001_c000~0x4001_ffff	SPI	
0x4002_0000~0x4002_3fff	I2C	
0x4002_4000~0x4002_7fff	I2S	
0x4002_8000~0x4002_bfff	PWM	
0x4002_c000~0x4002_ffff	GPIO	
0x4003_0000~0x4003_3fff	ADC	
0x4003_4000~0x4003_7fff	SDIO	
0x4003_8000~0x4003_bfff	RTC	

5.3 Boot Control

MODE	GPIO10	GPIO11	GPIO12	description
Normal Boot	1	1	1	All the GPIOs are PULL-UP inside.
UART Boot	0	1	0	CPU will wait data from UART to program flash.

Note: pull-up voltage should be compatible to VDDIO.

5.4 Interrupt Controller

NVIC inside CM4 is in charge of interrupts. There are 34 interrupts in whole system:

index	name	description
33	pa_otd_intr	PA over temperature
32	i2s_intr	I2S
31	gpio_intr	GPIO
30	spim2_intr	SPI mater 2
29	uart1_intr	UART1
28	uart0_intr	UART0
27	spis_intr	SPI slave
26	spim_intr	SPI master

25	i2c1_intr	I2C1
24	i2c0_intr	I2C0
23	timer_intr	Timer
22	adcc_intr	ADC control
21:16	cm4_intr[5:0]	CM4F exception: FPIXC FPOFC FPUFC FPIOC FPDZC FPIDC
15:7	sdio_intr[8:0]	SDIO
6	qspi_intr	QSPI
5	dmac_intr	DMA
4	mac_intr	WIFI MAC
3	idle	just a wakeup source, not interrupt
2	rtc_intr	real time circute
1	ext_intr	external interrupt, can also wakeup system
0	wdt_intr	watch dog

5.5 Timer

The Timer block contains 4 timer modules that are software controlled, programmable and can be used for various tasks. They are 24-bit general purpose timer with a PWM output capability. Timer clock frequency can vary from 1/2 to 1/256 of APB clock frequency.

5.6 DMA

The DMA controller has three Direct Memory Access (DMA) channels for fast data transfers from/to peripherals to/from any on-chip RAMs. The DMA controller off-loads the ARM interrupt rate if an interrupt is given after a number of transfers. 15 peripherals DMA requests are multiplexed on the 3 available channels, to increase utilization of the DMA service throughout the system.

index	request name
14	AUXADC
13	i2c1 rx
12	i2c1 tx
11	i2c0 rx
10	i2c0 tx
9	spis rx
8	spis tx
7	spim rx
6	spim tx
5	uart1 rx

4	uart1 tx
3	uart0 rx
2	uart0 tx
1	qspi tx
0	qspi rx

5.7 WDT

The watchdog timer is a 24-bit timer that can be used to detect an unexpected execution sequence caused by a software run-away and can generate a full system reset or an interrupt.

5.8 System Control Unit

There two system control unit in AWO and CMP domain respectively.

5.8.1 AWO system control

external interrupt control

- PMU configuration
- PAD configuration (such as pull-down/up, drive strength, hi-performance)
- SYSPLL configuration
- 32K RCO configuration
- 32K RCO real time calibration configuration
- Analog PMU configuration

5.8.2 CMP system control

All module clock gate and reset soft control
IO-mux configuration such as function selection
Save and restore of RF interface registers

6 System Interface

6.1 GPIO

LN8810 has totally 18 digital Input and Outputs (I/O) which can be set as general purpose input and output interfaces (GPIO).

- Two group IO, which are separately configurable
- Separate data registers and data direction registers for each signal
- Configurable hardware and software control for each signal, or for each bit of each signal
- Configurable interrupt mode for Port A
- Debounce logic with an external slow clock to debounce interrupts
- Generate single hi-active interrupts

6.2 SDIO

The SDIO is compatible with the SD memory card, which includes mechanical, electrical, power, signaling and software. It supports the standard SDIO version 2.0. It provides high-speed data I/O with low power consumption for mobile electronic devices. The main features are as follows:

- DMA supported
- The clock is up to 40MHz.
- Supported 1-bit,4-bit data mode
- Minimal or no modification to SD Physical bus
- Minimal change to memory driver software
- Extended physical form factor available for specialized applications
- Multi-function support including multiple I/O and combined I/O and memory
- Up to 7 I/O functions plus one memory supported on one card.
- Initialization Voltage: 2.0 to 3.6V
- Operational Voltage range:
 - Standard: 2.7-3.6V
 - Minimal: 3.1-3.5V

6.3 UART

The UART is compliant to the industry-standard 16550 and is used for serial communication with a peripheral, modem (data carrier equipment, DCE) or data set. Data is written from a master (CPU) over the APB bus to the UART and it is converted to serial form and transmitted to the destination device. Serial data is also received by the UART and stored for the master (CPU) to read back.

There are 2 UART in system, UART0 and UART1, the common feature of both are:

- 16 bytes Transmit and receive FIFOs.
- DMA supported
- Shadow registers to reduce software overhead
- Transmitter Holding Register Empty (THRE) interrupt mode
- Functionality based on the industry standard of 16550;
- Programmable character properties, such as number of data bits per character (5-8), optional
- parity bit (with odd or even select) and number of stop bits (1, 1.5 or 2)
- Line break generation and detection
- Prioritized interrupt identification
- Programmable serial data baud rate as calculated by the following: $\text{baud rate} = (\text{serial clock frequency})/(\text{divisor})$
- Furtherly, UART0 has exclusive function below:
- Hardware flow control support (CTS/RTS)
- IrDA 1.0 SIR mode supporting low power mode.

6.4 SPI Flash Controller

The Quad SPI Controller (QSPIC) provides a low pin count interface to FLASH memory devices. The QSPIC supports the standard Serial Peripheral Interface (SPI) and a high performance Dual/Quad SPI Interface. The QSPIC gives the ability to read data from a quad FLASH memory, transparently through the SPI bus. This Execute In Place (XIP) feature combined with the CPU cache, provides comparable performance to executing code from standard parallel FLASH. In this case the QSPIC generates all the control signals for the SPI bus that are needed to read data from the serial FLASH memory. Additionally, software can easily control the serial FLASH memory via a memory mapped register file which is contained in the QSPIC. All instructions supported by the FLASH memory, can be programmed using the above register file.

Programmable delay on the sample time of the received serial data bit (RXD), enables programmable control of routing delays resulting as high as 80MHz serial data-bit rates.

6.5 SPI

The SPI interfaces enable full duplex synchronous communication between devices. They support bi-directional bus with fast data transfers. There are three independent SPI interfaces: SPIM as four wire master, SPIM2 as a three-wire master, and SPIS as slave.

6.5.1 SPIM:

- Up to 3 individual chip-select signals
- 4-wire serial interface: SCK CSn SDI SDO
- 16 words Tx and Rx FIFO
- DMA interface supported
- Dynamic control of the serial bit rate of the data transfer, maximum frequency of the bit-rate clock (sclk_out) is one-half the frequency of APBCLK
- Programmable delay on the sample time of the received serial data bit

6.5.2 SPIM2

It is a simply 3-wire SPI master interface, and mainly used to config some external chip's registers. There are no FIFO, no DMA interface.

6.5.3 SPIS

- 4-wire serial interface: SCK CSn SDI SDO
- 16 words Tx and Rx FIFO
- DMA interface supported
- maximum frequency of the bit-rate clock (sclk_out) is 1/4 of the frequency of APBCLK

6.6 I2C

The I2C is a programmable control bus that provides support for the communications link between Integrated Circuits in a system. It is a simple two-wire bus with a software-defined protocol. Two identical I2C controllers are instantiated in the system, namely I2C0 and I2C1.

- Two-wire I2C serial interface consists of a serial data line (SDA) and a serial clock (SCL)
- Standard and Fast mode support (up to 400 kb/s)
- Clock synchronization
- 16 deep transmit/receive 11-bit wide FIFOs
- Master OR slave I2C operation

- 7 or 10-bit addressing
- 7 or 10-bit combined format transfers
- Bulk transmit mode
- Default slave address of 0x055
- Interrupt or polled-mode operation
- Handles Bit and Byte waiting at both bus speeds
- Programmable SDA hold time
- DMA request signals

6.7 I2S

There is one I2S interface used as slave. I2S interfaces are mainly used in applications such as data collection, processing, and transmission of audio data, as well as the input and output of serial data.

- I2S transmitter and/or receiver based on the Philips I2S serial protocol
- 2 channels for both transmitter and receiver
- Full duplex communication
- Asynchronous clocking of APB bus and I2S sclk
- Slave mode of operation
- Audio data resolutions of 12, 16, 20, 24, and 32 bits
- External sclk gating and enable signals
- Support for programmable DMA registers
- Programmable FIFO thresholds

6.8 PWM

There are 4 PWMs inside Timer, which has 24-bit width counter and using its own timer clock, another 4 PWMs are designed using 20-bit counter and APB clock. Duration of low level and high level are needed to set, and this determines the duty cycle of PWM waveform. These 8 PWMs can be flexibly configured to specific GPIOs with fullmux function.

6.9 Multi-channel input ADC

Linear Successive Approximation algorithm with 8-, 10-, or 12-bit resolution

- Up to 4 external analog inputs, external pin inputs, and 2 internal analog inputs including internal bandgap, temperature sensor, and references
- Output formatted in 8-, 10-, or 12-bit right-justified unsigned format
- Single or Continuous Conversion (automatic return to idle after single conversion)

Support up to eight result FIFO with selectable FIFO depth

- Configurable sample time and conversion speed/power
- Conversion complete flag and interrupt
- Input clock selectable from up to four sources
- Operation in Wait or Stop modes for lower noise operation

- Asynchronous clock source for lower noise operation
- Selectable asynchronous hardware conversion trigger
- Automatic compare with interrupt for less-than, or greater-than or equal-to, programmable value

7 Electrical Specifications

7.1 Absolute Maximum Ratings

Table 4. Absolute Maximum Ratings

Symbol	Min.	Max.	Unit
Supply Voltage	0	3.6	V
Digital IO & VDDIO	-0.3	VDDPMU+0.3	V
Storage Temperature	-40	120	°C

7.2 Recommended Conditions

Table 5. Operation Conditions

Symbol	Symbol	Min.	Typ.	Max.	Unit
Supply Voltage	VDD PMU	2.5	3.3	3.6	V
	DVDD*		1.1		V
	VDDIO*	1.8	2.5	3.3	V
Temperature		-40	25	85	°C

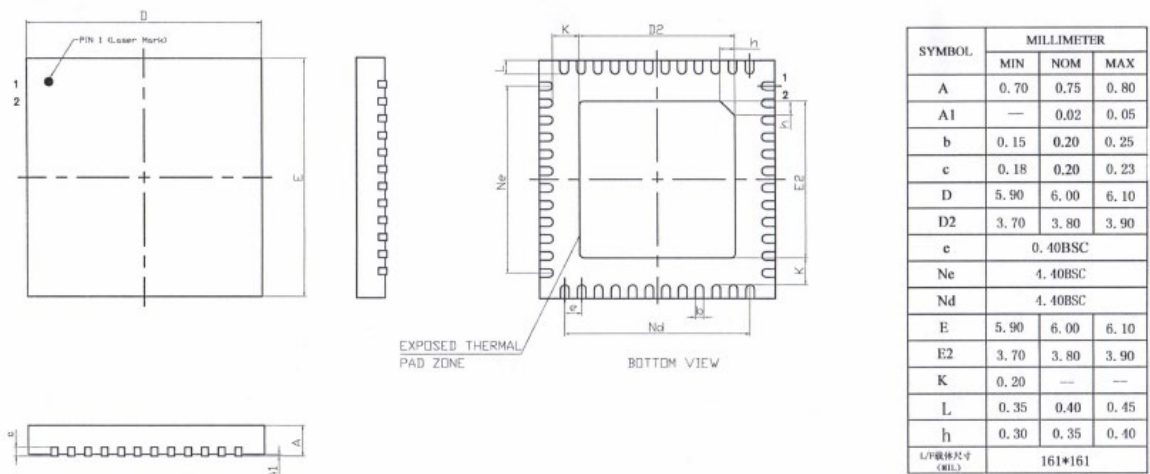
7.3 Logic Characteristics

Table 6. Logic IO Electrical Characteristics

Characteristic	Min.	Typ.	Max.	Unit
Input low voltage V_{IL}	-0.3	0	0.6	V
Input high voltage V_{IH}	VDDIO-0.6	VDDIO	VDDIO+0.3	V
Output low voltage V_{OL}	-0.45	0	0.45	V
Output high voltage V_{OH}	VDDIO-0.5	VDDIO	VDDIO+0.5	V

8 Package Information

The QFN48 package dimension is shown as below



9 Recommended Reflow Profile

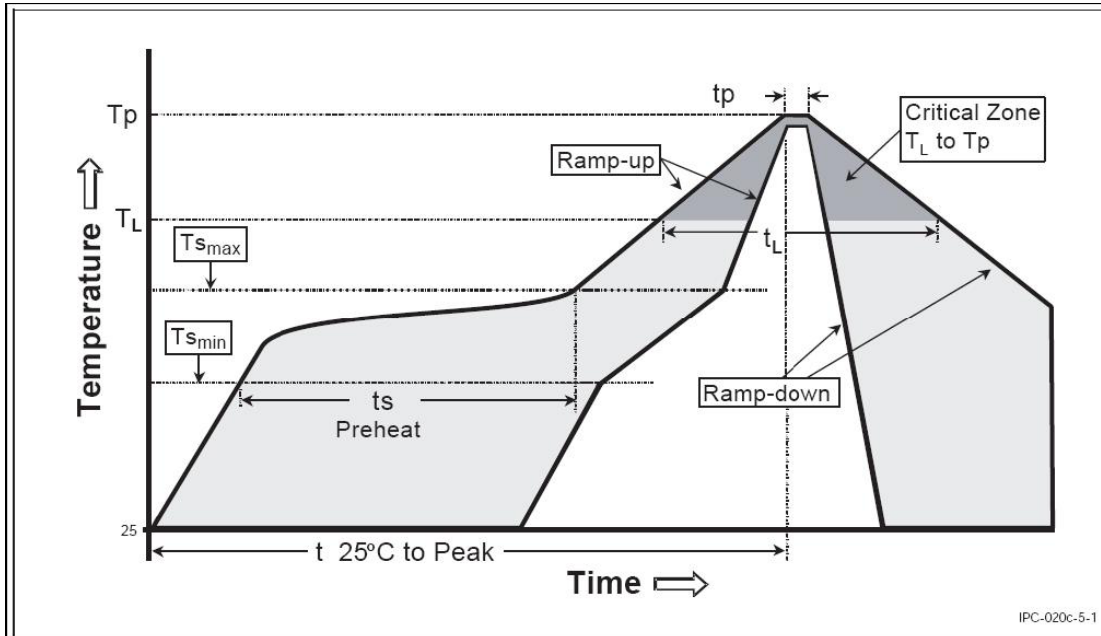


Figure.9-1 Classification Reflow Profile

Table 9-1 Classification Reflow Profiles

Profile Feature	Sn-Pb Eutectic Assembly	Pb-Free Assembly
Average Ramp-Up Rate (TSmax to Tp)	3 °C/second max.	3 °C/second max.
Preheat		
-Temperature Min (T _{min})	100 °C	150 °C
-Temperature Max (T _{max})	100 °C	200 °C
-Time (t _{min} to t _{max})	60-120 seconds	60-180 seconds
Time maintained above:		
-Temperature (T _L)	183 °C	217°C
-Time (t _L)	60-150seconds	60-150 seconds
Peak /Classification Temperature(T _p)	See Table 11-2	See Table 11-3
Time within 5 oC of actual Peak Temperature (t _p)	10-30 seconds	20-40 seconds
Ramp-Down Rate	6 °C/second max.	6 °C/seconds max.
Time 25 oC to Peak Temperature	6 minutes max.	8 minutes max.

Table 9-2 Sn-Pb Eutectic Process – Package Peak Reflow Temperatures

Package Thickness	Volume mm3 <350	Volume mm3 ≥350
<2.5mm	240 + 0/-5 °C	225 + 0/-5 °C
≥2.5mm	225 + 0/-5 °C	225 + 0/-5 °C

Table 9-3 Pb-free Process – Package Classification Reflow Temperatures

Package Thickness	Volume mm3 <350	Volume mm3 350-2000	Volume mm3 >2000
<1.6mm	260 + 0 °C *	260 + 0 °C *	260 + 0 °C *
1.6mm – 2.5mm	260 + 0 °C *	250 + 0 °C *	245 + 0 °C *
≥2.5mm	250 + 0 °C *	245 + 0 °C *	245 + 0 °C *
*Tolerance : The device manufacturer/supplier shall assure process compatibility up to and including the stated classification temperature(this mean Peak reflow temperature + 0 °C. For example 260+ 0 °C) at the rated MSL Level.			

Note 1: All temperature on topside of the package. Measured on the package body surface.

Note 2: The profiling tolerance is + 0 °C, - X °C (based on machine variation capability) whatever is required to control the profile process but at no time will it exceed – 5 °C. The producer assures process compatibility at the peak reflow profile temperatures defined in Table 8-3.

Note 3: Package volume excludes external terminals (balls, bumps, lands, leads) and/or non-integral heat sinks.

Note 4: The maximum component temperature reached during reflow depends on package the thickness and volume. The use of convection reflow processes reduces the thermal gradients between packages. However, thermal gradients due to differences in thermal mass of SMD package existed.

Note 5: Components intended for use in a “lead-free” assembly process shall be evaluated using the “lead free” classification temperatures and profiles defined in Table8-1, 8-2, 8-3.

10 RoHS Compliant

This product does not contain lead, mercury, cadmium, hexavalent chromium, polybrominated biphenyls (PBB) or poly brominated diphenyl ethers (PBDE), so it is therefore compliant with RoHS.

11 ESD Precautions

ESD protection circuitry is contended in this device, but special handling precautions are required.

12 Disclaimer

The information provided here is reliable; it assumes that no reliability for inaccuracies and omissions, and no reliability for the use of this information and all such information should entirely be the user's own risk.

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